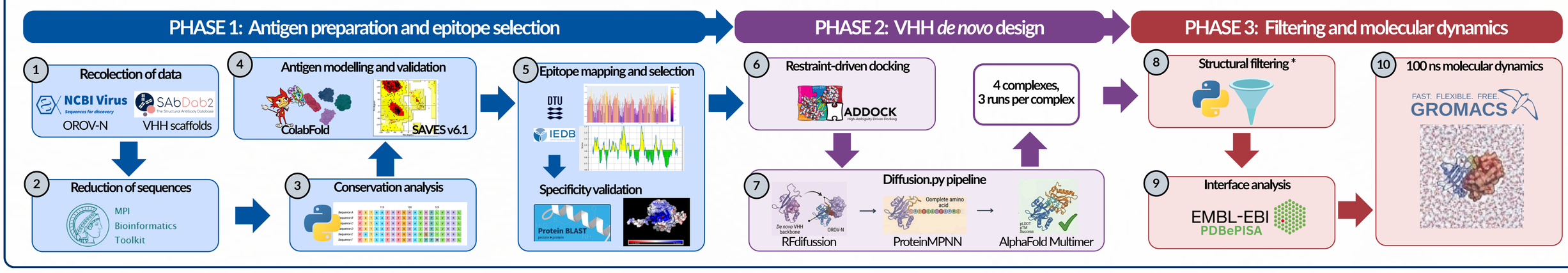
De novo computational design of epitope-specific nanobodies against the *Orthobunyavirus oropoucheense* nucleocapsid for point-of-care diagnostics

Alcántara-Ríos, C.[1] ; Hernandez, A.[1] ; Sandoval, G.[1]
[1] Faculty of Biological Sciences, Universidad Nacional Mayor de San Marcos, Peru

Background

Orthobunyavirus oropoucheense (OROV) causes Oropouche fever, an acute febrile arboviral disease transmitted mainly by *Culicoides paraensis* midges, with documented neurological complications and recent evidence of vertical transmission. The 2023–2025 epidemic exceeded 23,000 cases across the Americas, exposing critical diagnostic gaps in endemic rural regions. RT-qPCR remains the gold standard but requires infrastructure unavailable at the point of care (POC). No epitope-directed lateral flow reagent currently exists for the Oropouche species complex. Nanobodies (VHH) (~15 kDa), thermostable, bacterially expressible single-domain antibody fragments, are ideal POC diagnostic reagents. Traditional discovery requires camelid immunization (slow, costly, animal-dependent). Deep learning tools (RFdiffusion, ProteinMPNN) now enable fully *de novo*, epitope-directed nanobody design without immunization, targeting conserved regions of the OROV nucleocapsid (OROV-N) to maximize serogroup coverage and minimize cross-reactivity with co-circulating arboviruses.

Methodology



Results

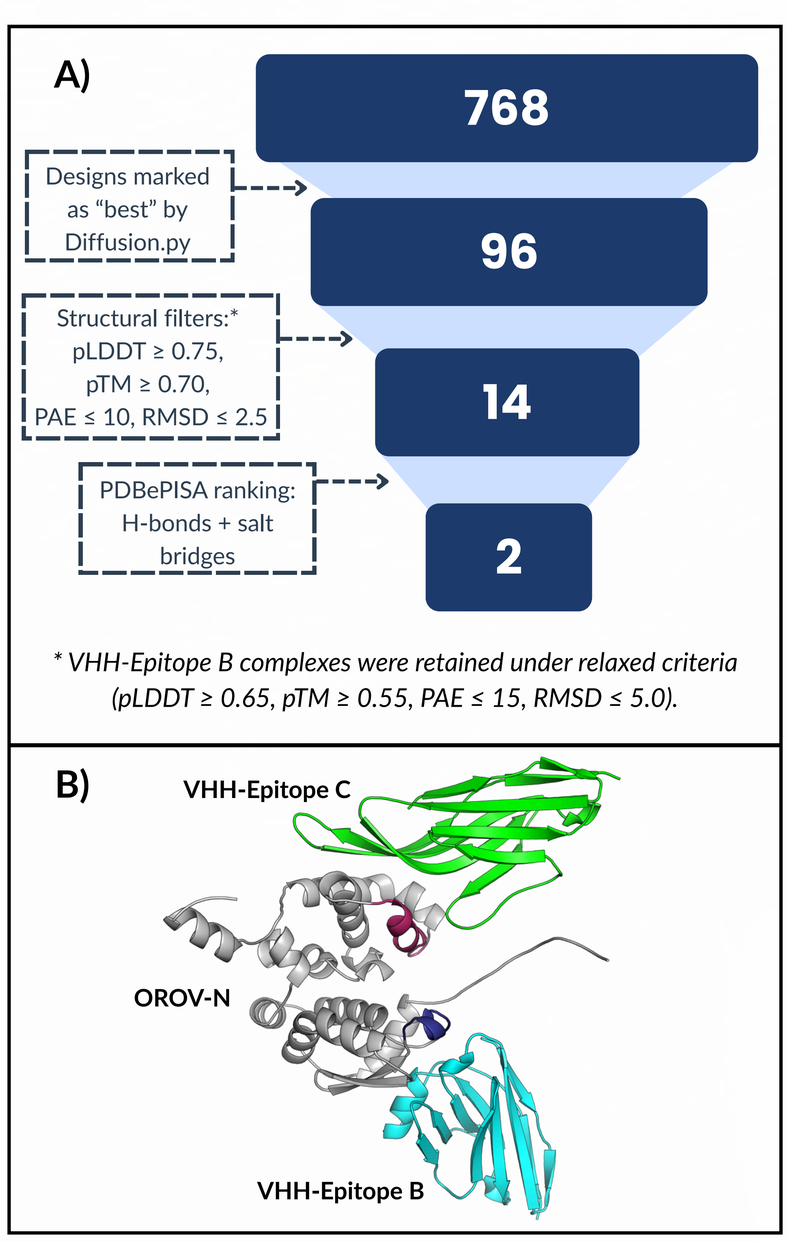


Figure 2. Complex selection process. (A) Structural filters applied to the directed mutagenesis results. (B) Formation of the complex between the 2 best VHH candidates of each epitope.

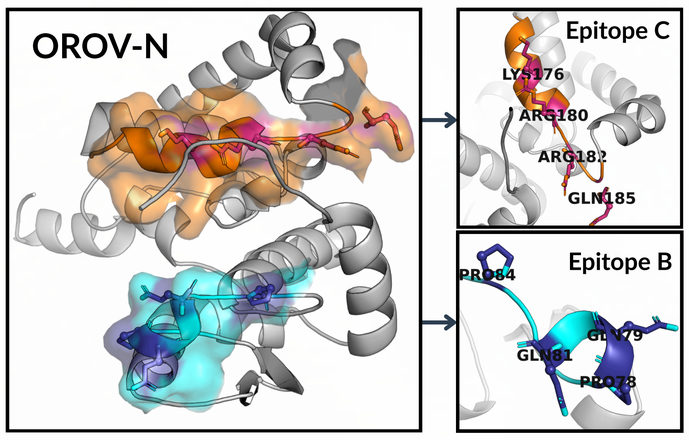


Figure 1. Structural localization of the two selected epitopes on the OROV-N. Epitope C: a surface-exposed region with low RNA-binding interference (Shannon entropy (*H*) = 0.01667; SASA = 0.80–170.70 Å² per residue). Epitope B: a partially buried region with moderate RNA-binding interference (*H* = 0.01290; SASA = 23.99–117.93 Å² per residue).

Supplementary tables

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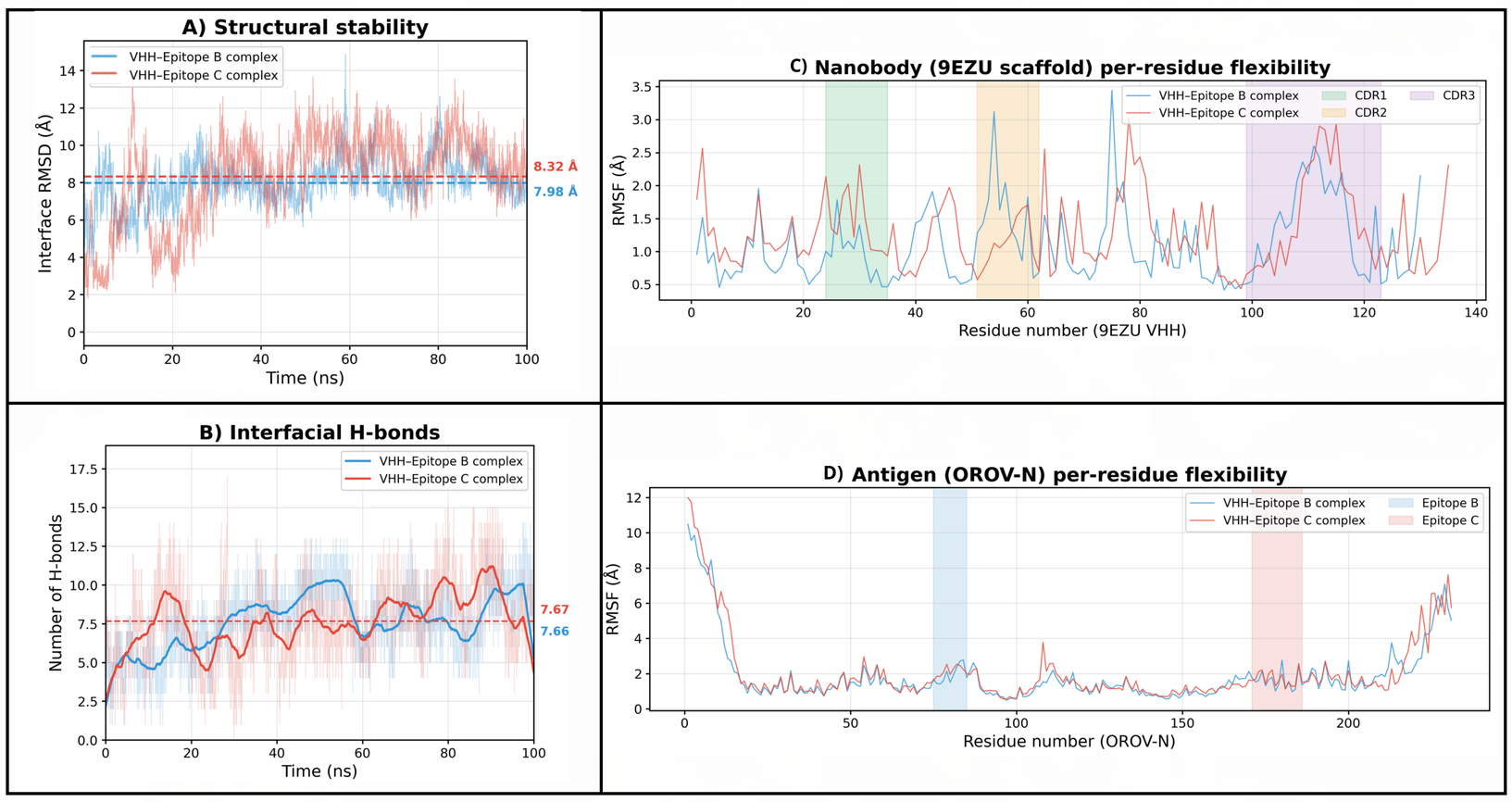


Figure 3. Molecular dynamics simulation analysis of VHH-epitope complexes over 100 ns. (A) Interface RMSD (B) Interfacial H-Bonds (C) Per-Residue RMSF of the nanobody scaffold (9EZU VHH) (D) Per- residue RMSF of the antigen (OROV-N)

Conclusions

- This fully computational, epitope-directed framework enables a generalizable, epitope-directed framework for rapid, low-cost nanobody design against emerging arboviruses, supporting the accelerated development of species-specific POC lateral flow assays for resource-limited settings in Latin America.
- Successfully designed 2 epitope-specific nanobodies *de novo* showing high sequence conservation across the Oropouche complex and no cross-reactivity with co-circulating arboviruses (Zika, Dengue, Chikungunya, and Mayaro).
- Future directions include the experimental validation (in vitro) of these designed nanobodies and the expansion of the screening library to optimize binding affinities for point-of-care diagnostic integration

References

Acknowledgements

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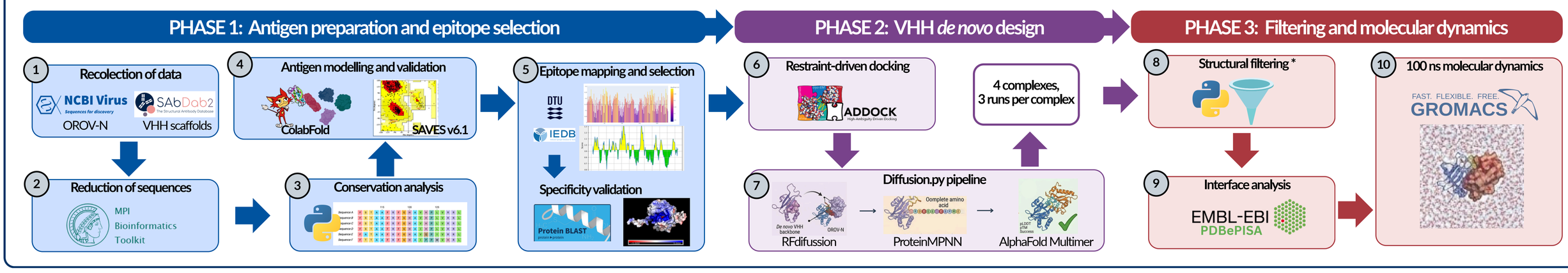
Addressing the Oropouche diagnostic gap: *in silico* design of pan-specific nanobodies against the nucleocapsid protein for point-of-care applications

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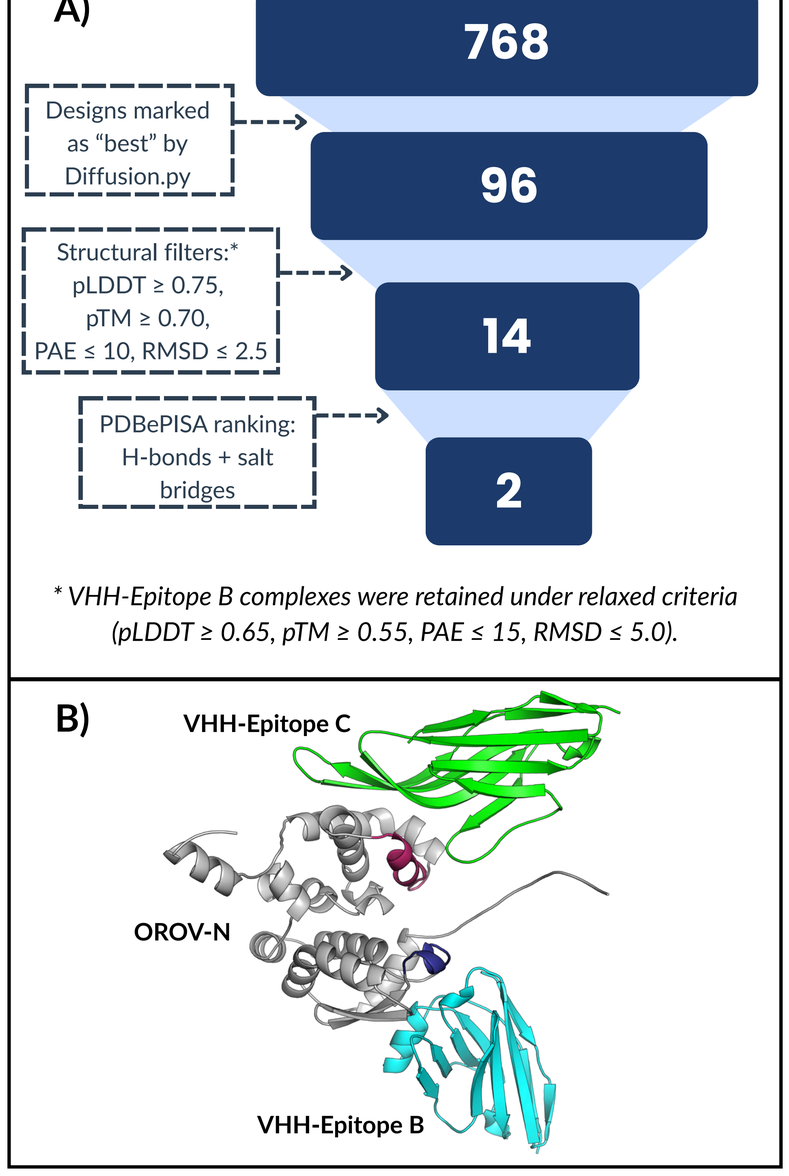


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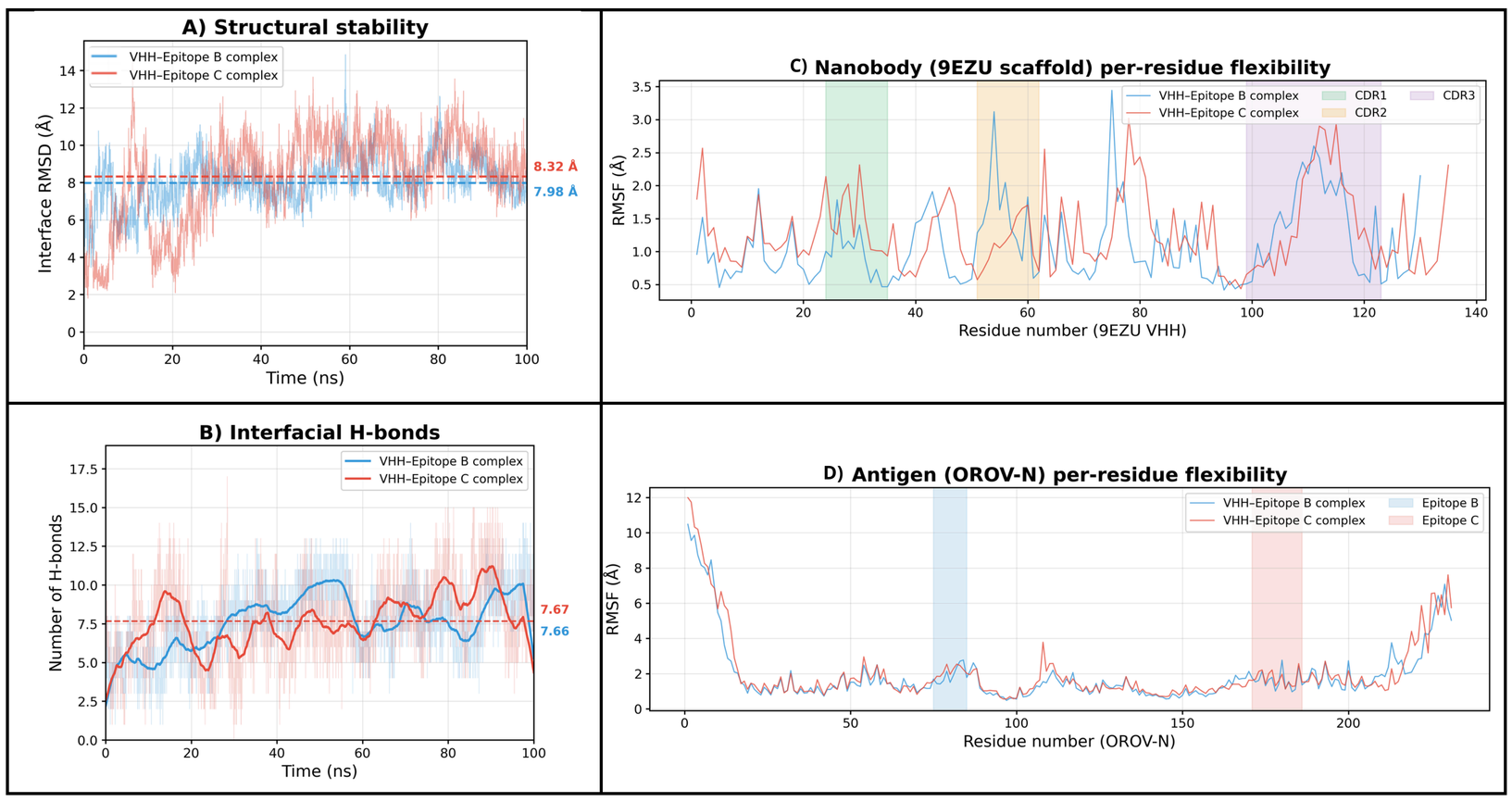


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